NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-39016/45B
20 JULY 1988
SUPERSEDING
MIL-R-39016/45A
13 August 1985

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT, LOW LEVEL TO 2.0 AMPERES (LATCHING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-R-39016.

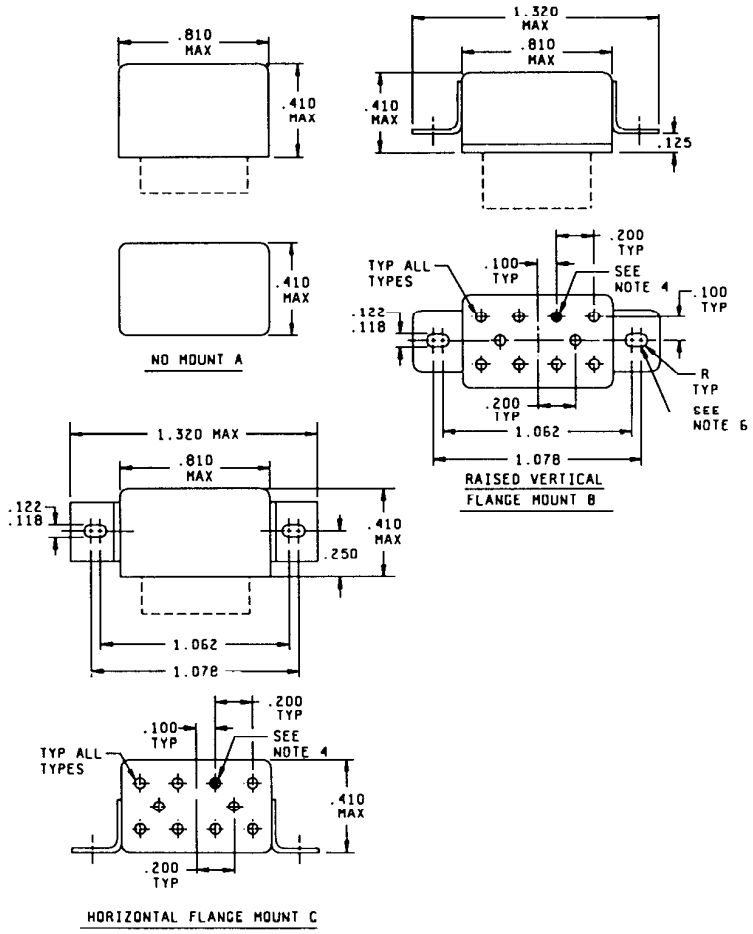
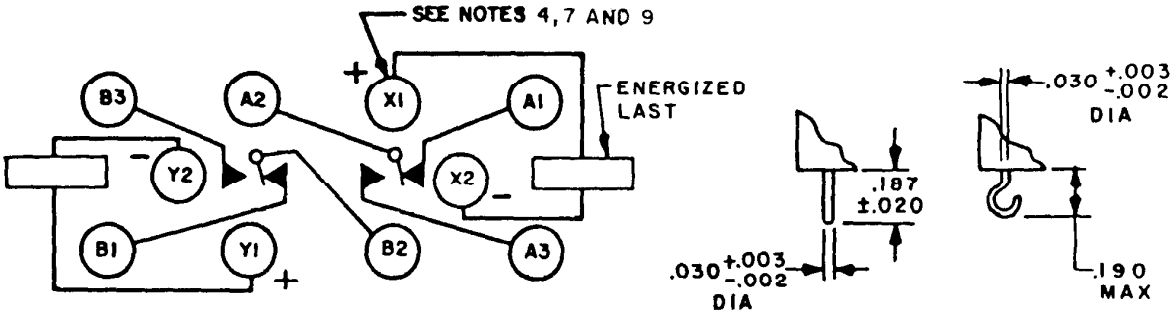


FIGURE 1. Dimensions and configuration.

(B) denotes changes

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(B) TERMINAL VIEW CIRCUIT DIAGRAM

PRINTED WIRE (PW) LEAD

SOLDER LUG (SL) LEAD

Inches	mm	Inches	mm
.002	0.05	.250	6.35
.003	0.08	.400	10.16
.015	0.38	.410	10.41
.020	0.51	.412	10.46
.030	0.76	.455	11.45
.100	2.54	.600	15.24
.120	3.05	.810	20.57
.125	3.18	1.062	26.97
.187	4.75	1.078	27.38
.190	4.83	1.320	33.53
.200	5.08	2.950	74.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal indicated shall be identified by a contrasting bead. Relays shall have (+) and (-) signs placed on the circuit diagram as shown.
5. Mounting screw head clearances based on the use of No. 4 fillister head screws.
6. Coil symbol optional in accordance wtih MIL-STD-1285.
7. When relay enclosure has side mounting hardware, the contrasting header bead shall be located on the opposite side from the mounting hardware.
8. Mounting surface finish shall be compatible with aluminum (dura-aluminum type) as defined by the compatible couples table of MIL-R-39016.
9. Energizing the indicated coil with proper polarity and voltage shall cause the relay contacts to assume the position shown.
10. (B) Terminal numbers in circuit diagram are for reference only. Numbers do not appear on relay.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive: 2.0 amperes at 28 V dc; 0.150 ampere at 115 V ac, 60 and 400 Hz. 1/
Inductive: 0.75 ampere at 28 V dc with 200 millihenries inductance.

Lamp: 0.160 ampere at 28 V dc.

Low level: 10 to 50 microamperes at 10 to 50 millivolts dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

(B) *Initial: 0.050 ohm maximum.*

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: .100 ohm maximum.

Low level:

During life: 50 ohms maximum.

(B) After life: 0.150 ohm maximum.

Intermediate current:

During intermediate current: 3 ohms maximum.

(B) After intermediate current: 0.100 ohm maximum.

Contact bounce (applicable to failure rate level "L"): 2.0 milliseconds (ms) maximum.

Contact stabilization time (applicable to failure rate levels "M", "P", and "R"): 2.5 ms maximum.

Overload (high level only): Two times rated current.

(B) Neutral screen: Applicable.

(B) COIL DATA: See table I.

Duty rating: Continuous.

Operate time: 5 ms maximum over temperature range with rated coil voltage.

Release time: Not applicable.

Coil life: Not applicable.

ELECTRICAL DATA: 2/

Insulation resistance: 1,000 megohms minimum, except the resistance between coil and case at high temperature shall be 500 megohms or greater.

1/ For qualification, testing of 400 Hz loads not required.

2/ (For qualification and group C tests only). During insulation resistance and dielectric withstanding voltage tests, the relay shall be latched in each of the two positions.

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Dielectric withstanding voltage:	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts	1,000	350 All terminals to case
Between case, frame, or enclosure and coils	500	
Between all contacts and coils	1,000	
Between open contacts in the latching and reset positions	500	
Between coils	500	
Between contact poles	1,000	

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204 (except frequency shall be 10 to 2,500 Hz). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

(B) Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts (applicable to qualification and group C testing only).

Shock (specified pulse): MIL-STD-202, method 213, test condition C (100 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt spray (corrosion): Applicable.

PHYSICAL DATA:

Terminals: See figure 1 and table I.

Terminal strength: 3 ±0.3 pounds pull.

(B) Solderability: Applicable.

Dimensions and configurations: See figure 1.

Weight: 13 grams (0.46 ounce) maximum.

Seal: Hermetic.

Identification marking (full): Applicable.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles.

(B) Low level: 100,000 cycles plus 900,000 cycles mechanical life.

PART NUMBER: M39016/45-(dash number from table I and suffix letter designating failure rate level).

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Ⓑ TABLE I. Dash numbers and characteristics. 1/ 2/

Dash numbers 3/					At 25°C			Over temperature range	
Solder lug	Printed wire (PW) .187 ±.020	Mount	Coil voltage (V dc) 4/		Coil resistance (ohms ±10%)	Specified pickup (latch/reset) value (voltage) (V dc)		Specified pick-up (latch/reset) value (voltage) (V dc)	
			Rated	Max		Max	Min	Max	Min
001	002	A	5.0	6.7	45	2.7	1.6	3.8	1.0
003	004	B	5.0	6.7	45	2.7	1.6	3.8	1.0
005	006	C	5.0	6.7	45	2.7	1.6	3.8	1.0
007	008	A	6.0	8.0	63	3.25	2.0	4.5	1.3
009	010	B	6.0	8.0	63	3.25	2.0	4.5	1.3
011	012	C	6.0	8.0	63	3.25	2.0	4.5	1.3
013	014	A	12.0	16.0	254	6.5	4.0	9.0	2.6
015	016	B	12.0	16.0	254	6.5	4.0	9.0	2.6
017	018	C	12.0	16.0	254	6.5	4.0	9.0	2.6
019	020	A	26.5	32.0	1000	13.0	8.0	18.0	5.2
021	022	B	26.5	32.0	1000	13.0	8.0	18.0	5.2
023	024	C	26.5	32.0	1000	13.0	8.0	18.0	5.2
025	026	A	48.0	64.0	3800	26.0	16.0	36.0	10.4
027	028	B	48.0	64.0	3800	26.0	16.0	36.0	10.4
029	030	C	48.0	64.0	3800	26.0	16.0	36.0	10.4

1/ WARNING: When latching relays are installed in equipment, the latch and reset coils should not be pulsed simultaneously. Coils should not be pulsed with less than the nominal coil voltage and the pulse width should be a minimum of three times the specified operate time of the relay. If these conditions are not followed, it is possible for the relay to be in the magnetically neutral position.

2/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 Vdc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

3/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 001L - - - - - - 012R.

4/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

Ⓑ TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
20 units plus 1 unsealed unit for level L at C = 0 1/	M39016/45-021	20 units plus 1 unsealed unit for level L at C = 0 1/
33 units plus 1 unsealed unit for level M at C = 0 1/		33 units plus 1 unsealed unit for level M at C = 0 1/
Qualification inspection as applicable		Qualification inspection as applicable
	M39016/45-010	2 units, qualification inspection table, group II, and shock, vibration, acceleration, terminal strength, and seal.
	M39016/45-003 M39016/45-009 M39016/45-015 M39016/45-027	2 units, qualification inspection table, group II.
		1 unsealed sample unit.

1/ The number of units required for qualification testing will be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection testing; the relay manufacturer shall preselect the sampling plan.

SUPERSESSION DATA: See table III.

Ⓑ TABLE III. Supersession data.

Inactive part no. M5757/76-	New part no. 1/ M39016/45-
001	008
002	014
003	020
004	026
013	009
014	015
015	021
016	027
017	011
018	017
019	023
020	029

1/ Complete part number shall contain suffix letter L, M, P, or R to designate failure rate level (see 3/ of table I). A part with any failure rate supersedes the applicable MIL-R-5757 part.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS
(B) Air Force - 11
DLA - ES

User activities:
Army - SM
Navy - MC, OS, SH
(B) Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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